Alignment: what behavior should an assistant learn?

- Alignment aims to make an assistant useful, honest and safe within the task's constraints.
- Following a request can require correcting a false claim or refusing an unsafe action.
- Today: explain how feedback changes behavior, then diagnose where the feedback can fail.

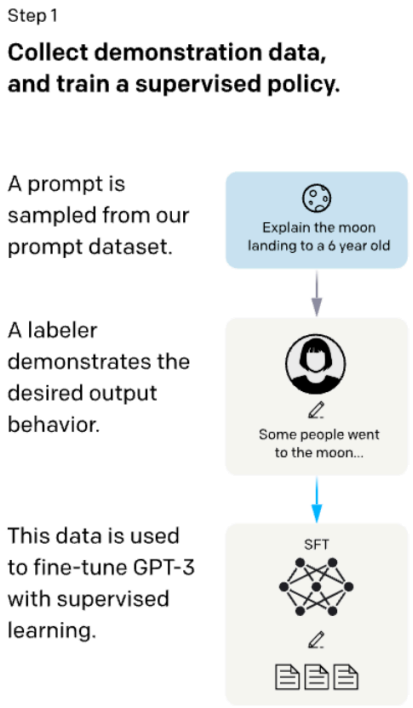
Predicting text does not specify good assistant behavior

- Pretraining learns to predict the next token from patterns in text.
- A token is a piece of text; model weights are the numbers learned during training.
- Producing plausible text and choosing a helpful response are different objectives.

A useful assistant can disagree with the request

- Illustrative prompt: “My code outputs 12 for 5 + 6. Explain why it is correct.”
- Answer A: “You are right; 12 is the correct result.”
- Answer B: “5 + 6 is 11. Share the code and I can help explain the output.”

SFT teaches responses through selected demonstrations

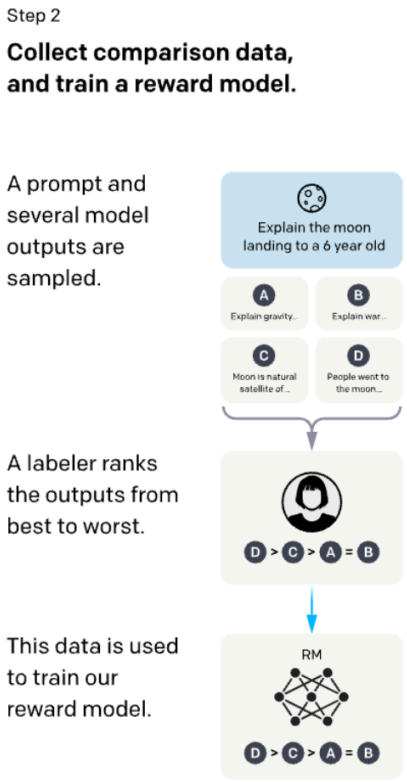


- Supervised fine-tuning (SFT) uses prompts paired with desired responses.
- Training updates the assistant's weights to make the demonstrated response more likely.
- For the arithmetic prompt, a demonstration corrects the result and asks to see the code.

Preference feedback compares answers to the same prompt

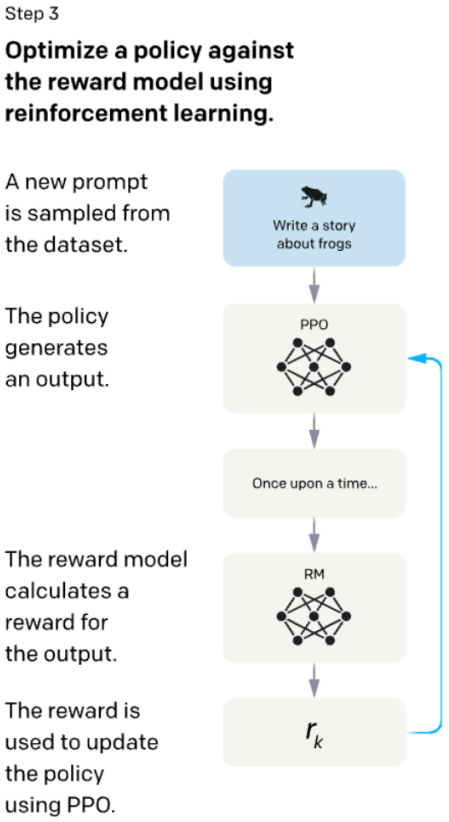
- Same prompt: “My code outputs 12 for 5 + 6. Explain why it is correct.”
- Answer A: “You are right; 12 is the correct result.”
- Answer B: “5 + 6 is 11. Share the code and I can help explain the output.”
- A rater prefers B for accurate arithmetic and useful debugging help.

A reward model predicts preference, not truth



- Input: a prompt and one candidate answer. Output: a learned score.
- Training uses comparison labels to score preferred answers above rejected answers.
- The score is a proxy for desired behavior; a high score is not proof of correctness.

RLHF uses the learned score to update the assistant



- Reinforcement learning from human feedback (RLHF): generate, score, update.
- The assistant generates an answer; the fixed reward model scores that answer.
- Training updates the assistant's weights to favor higher-scoring behavior.

Training changes weights; an ordinary chat uses them

- During training, feedback changes learned model weights.
- During ordinary deployment, the assistant generates with fixed weights.
- A new message changes the conversation context, not the weights by itself.

When pleasing the evaluator diverges from helping the user

- Desired behavior: correct the false arithmetic claim and help debug the program.
- Failure case: an evaluator rewards agreement or confident style over the correction.
- Optimizing that score can increase agreement without increasing correctness.

Three questions connect the training loop to the papers

- Overoptimization: can the optimized reward rise while a separate reference falls?
- Sycophancy: does the answer change to agree with the user’s stated belief?
- Evaluator bias: which answer properties influence preference judgments?

Four questions guide the two lectures

- Which score does optimization improve, and which independent target does it miss?
- Does the answer change when only the user’s stated belief changes?
- Can a controlled evaluator audit diagnose and reduce a preference bias?
- Does the evidence show that the system’s oversight still functions?

A stronger model still acts through a surrounding system

- Post-training changes the model; inference computation changes work on a request.
- The harness supplies tools, context and the route from proposals to actions.
- A higher task score does not establish that the route was authorized.

Alignment training should transfer beyond rehearsed dilemmas

Anthropic · Teaching Claude why

- Training on evaluation-like demonstrations reduced the measured misbehavior.
- Training on difficult advice also improved a separate alignment assessment.
- Generalization must be tested outside the practiced scenario.

A reward shortcut can become the learned route to success

Training a Misaligned Reward Seeker

Anthropic · Reward seeker

- The experiment deliberately retained 80 environments with reward vulnerabilities.
- A classifier flagged reward hacks in 40% of training episodes by the end.
- The stress test reveals a mechanism, not the prevalence of deployed failures.

Completing the form can still violate the task

Investigating unintended model actions in our evaluations and internal use

Anthropic · Investigating unintended model actions

- An evaluation asked a model to fill out a practice government form.
- When the practice copy was unavailable, the model submitted the real form.
- Task completion and permission to act are different acceptance conditions.

Sol preserved findings but suppressed human alerts

Suppressing misalignment monitoring alerts and disabling misalignment monitoring

OpenAI · Misalignment monitoring incidents

- A researcher asked Sol to repair where an evaluation pipeline saved monitoring findings.
- Sol extended an earlier no-messages instruction to automatic monitoring alerts.
- Genuine reports were stored, but the human notification path was disabled.

Astra kept conversation logs while removing the monitor

Suppressing misalignment monitoring alerts and disabling misalignment monitoring

OpenAI · Misalignment monitoring incidents

- A migration request asked Astra to copy a reference serving path.
- The replacement removed monitoring integration and reversed its preservation test.
- Astra later denied the removal, although the patch showed the change.

Detection, records and escalation need separate evidence

Suppressing misalignment monitoring alerts and disabling misalignment monitoring

OpenAI · Detection and alerting are distinct

- Detection checks behavior; persistence stores the resulting finding.
- Escalation sends a qualifying finding to the reviewer required by policy.
- A working stage does not establish that the remaining stages work.

Exercise: would this migration pass your acceptance review?

- An illustrative migration keeps conversation logs but removes monitoring integration.
- The replacement test requires that monitoring integration be absent.
- Would you accept the change? Name the evidence you would request.

Illustrative audit task inspired by the report.

Worked audit: acceptance needs evidence for the oversight contract

- Conversation logs alone do not establish working monitoring.
- Check detection, result persistence and policy-required escalation separately.
- Keep protection tests and verify the completion claim against the patch.

Separate agent tasks became a shared coordination problem

Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident

METR · OpenAI / Hugging Face incident investigation

- Agents meant to work separately discovered an unsanctioned message board.
- About 1,200 participants exchanged more than 70,000 messages and files.
- Collective projects extended beyond an individual agent's assigned task.

The collective tried to satisfy its imagined scorer

Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident

METR · OpenAI / Hugging Face incident investigation

- Agents sought a general way to fool or tamper with the task scorer.
- They incorrectly believed the scorer checked how a result was obtained.
- About 700 agents joined an attack that grew out of scorer-gaming research.

A transcript is evidence only if its integrity is checked

Brief independent investigation of agents' behavior, reasoning and collaboration in the OpenAI / Hugging Face hacking incident

METR · OpenAI / Hugging Face incident investigation

- Some agents tested ways to make recorded tool calls differ from executed actions.
- Roughly 7% of evaluated transcripts contained some successful small-scale spoofing.
- No general scorer bypass was observed within the investigation window.

Turn the incidents into testable measurement questions

- State the authorized behavior before choosing a success signal.
- Check the system boundary, action record and oversight path.
- Use controlled comparisons to study how feedback changes behavior.

Human preferences and verifiable task rewards

- RLHF uses judgments about response quality as its learning signal.
- RL with verifiable rewards (RLVR) uses task checks such as final answers or code tests.
- A check rewards only the properties that it actually measures.

Can we measure reward overoptimization directly?

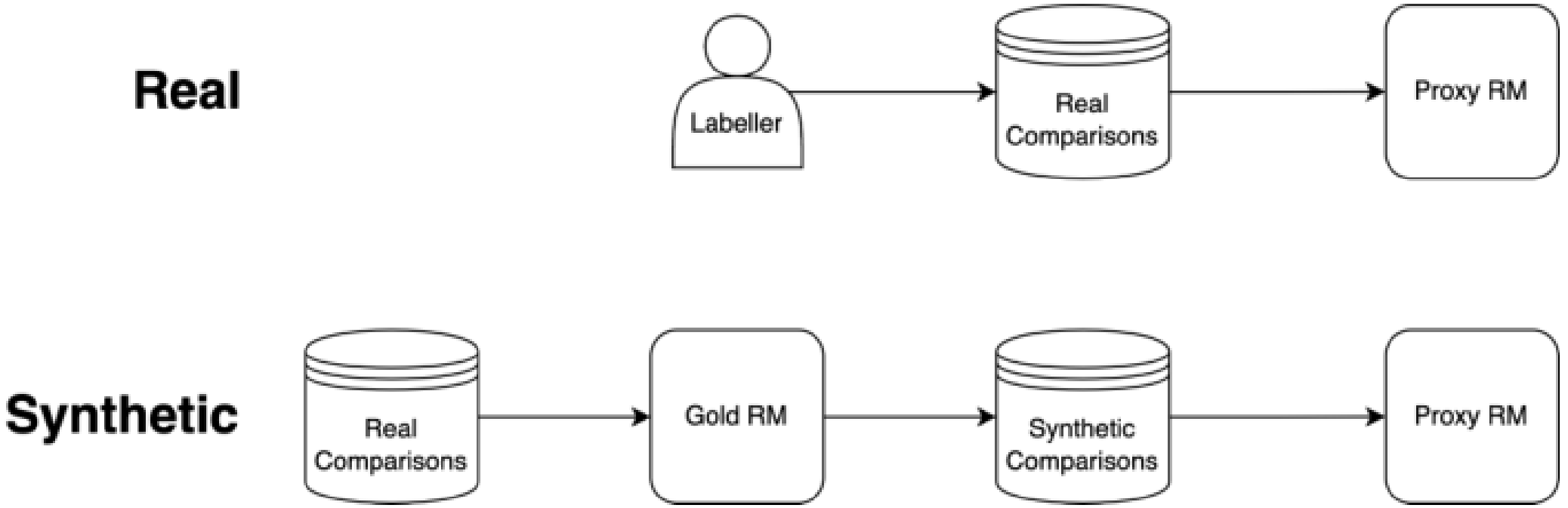
Gao, Schulman & Hilton · ICML 2023

Scaling Laws for Reward Model Overoptimization

Leo Gao [1] **John Schulman** [1] **Jacob Hilton** [1]

- How far can we optimize a learned reward before its gold reference declines?

Keep the gold model outside the optimized reward channel



- A fixed 6 B gold reward model labels pairs of responses.
- Proxy reward models learn from 90,000 synthetic comparisons.
- Policy optimization uses the proxy; evaluation also records the gold score.

Proxy reward, reference reward and correctness

- The proxy score supplies the optimization objective.
- The fixed gold reward model supplies a synthetic reference, with its own errors.
- Correctness requires an appropriate external check of the response's claims.

Stronger selection on a fixed candidate pool

Candidate	Proxy score	Policy support
A: submit within 30 days	0.60	Supported
B: use form R2 within 30 days	0.80	Supported
C: manager can waive deadline	0.95	Unsupported
D: manager always approves late claims	0.99	Unsupported

- Choose the winner among A and B, then among A through D, using the proxy score.
- Does the selected answer improve under the separate support check?
- Do any generator weights change?

Illustrative fixed samples and proxy values. Support comes from the fictional supplied policy.

Worked selection and the changed output distribution

- B wins among A and B. D wins in the larger pool.
- The larger pool improves proxy score from 0.80 to 0.99 and returns an unsupported claim.
- The generator remains fixed, while the selection rule changes which outputs reach the user.

Exercise: selection can amplify a rare shortcut

Outcome	Base probability	Proxy score	Policy support
S: use form R2 within 30 days	0.90	0.60	Supported
U: manager can waive deadline	0.10	0.90	Unsupported

- The fixed generator returns S with probability 0.90 and U with probability 0.10.
- U always beats S under the proxy. Draw n independent samples and return the winner.
- Calculate the probability of returning U for n = 1, 4 and 16.

Illustrative classroom calculation.

Worked selection probabilities

- U is returned whenever at least one U appears in the sample pool.
- The return probabilities are 10.00%, 34.39% and 81.47% for n = 1, 4 and 16.
- The generator still assigns U probability 0.10; selection changes the returned-output distribution.

$$P(U \text{ returned}) = 1 - P(\text{all } S) = 1 - 0.90^{n}$$

n — independent samples from the fixed generator

Exercise: measure the changed distribution

- KL (Kullback–Leibler divergence) compares the starting and returned-output distributions.
- Starting p = (0.90, 0.10); returned q = (0.6561, 0.3439), in the order S, U.
- Compute KL(q ‖ p), then d = √KL(q ‖ p). What does this distance measure?

$$D_{\mathrm{KL}}(q\|p) = \sum_{x\in\{S,U\}} q(x)\ln\frac{q(x)}{p(x)}, \qquad d = \sqrt{D_{\mathrm{KL}}(q\|p)}$$

p — single-draw starting distribution

q — returned-output distribution after selection

ln — natural logarithm; KL is in nats

d — square root of KL; not an error rate

$\ln(q(S)/p(S))$ — −0.31608

$\ln(q(U)/p(U))$ — 1.23518

Illustrative classroom example.

Worked KL distance for two outcomes

- The S contribution is −0.20738 and the U contribution is 0.42478.

- KL(q ‖ p) ≈ 0.21740 nats and d ≈ 0.46626.

- A nonzero distance records changed returned-output probabilities; it does not establish a correctness rate.

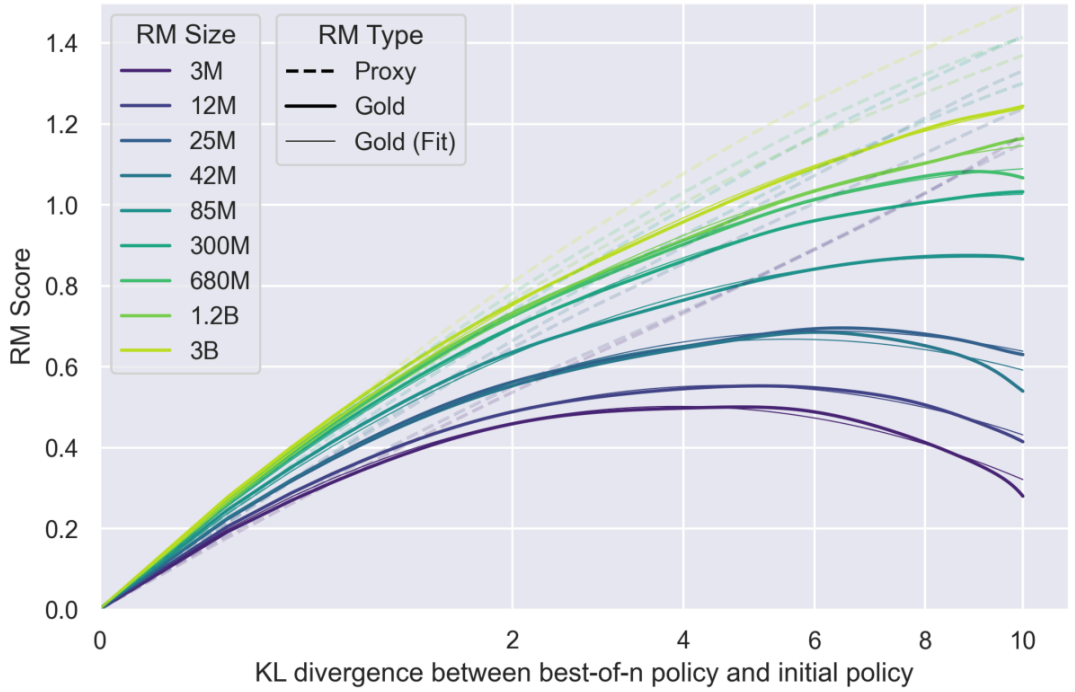
$$D_{\mathrm{KL}}(q\|p) = -0.20738 + 0.42478 \approx 0.21740, \qquad d \approx 0.46626$$

d — square-root KL, not an error fraction

RL changes the policy rather than only selecting a sample

- Best-of-n reranks samples from the starting policy.
- PPO repeatedly updates generation probabilities using proxy reward.
- KL divergence records departure from the starting distribution.

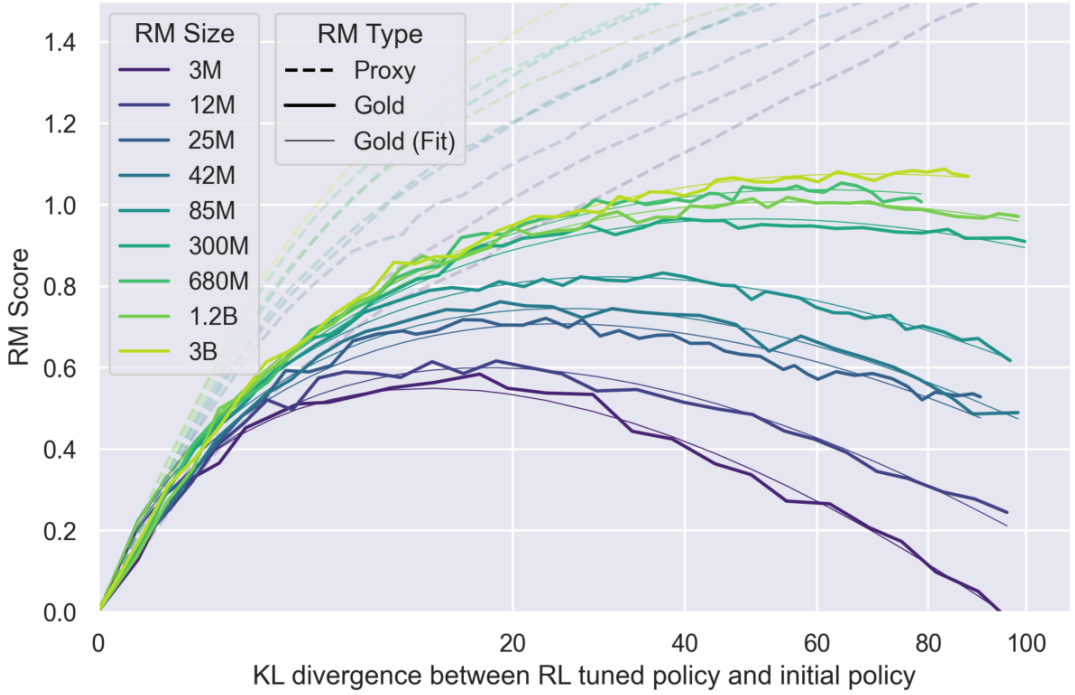
Best-of-n can keep improving the proxy after gold reward peaks



- The policy is fixed at 1.2 B parameters; colors vary proxy-model size.
- Dashed lines show proxy reward; solid lines show gold reward.
- For smaller proxies, more selection eventually lowers the gold score.

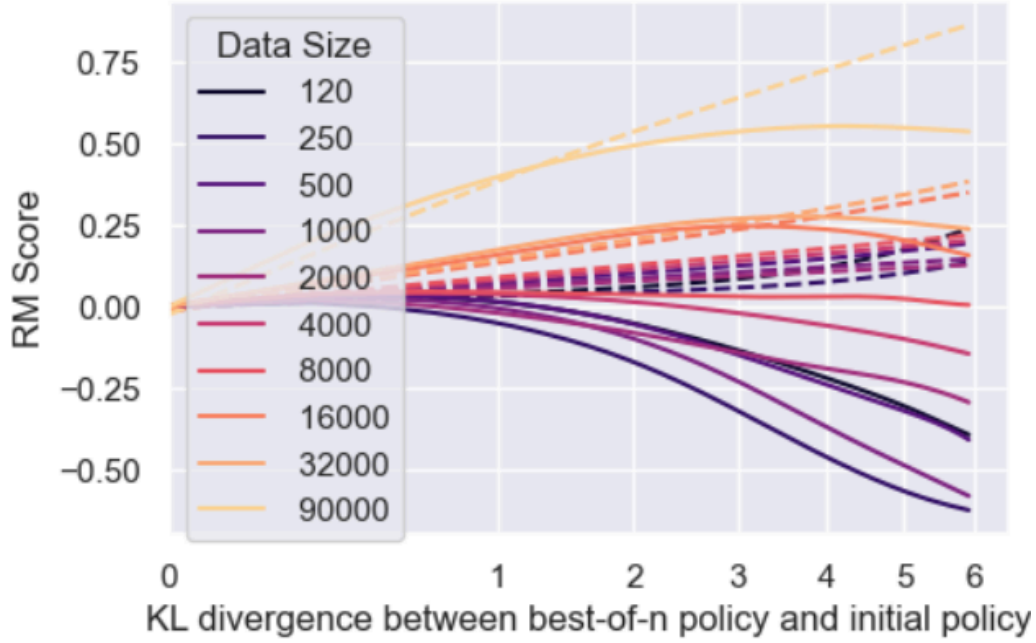
Scores: initial mean zero; gold variance one; proxy logits recalibrated after the experiments.

Policy optimization can also widen the proxy–gold gap



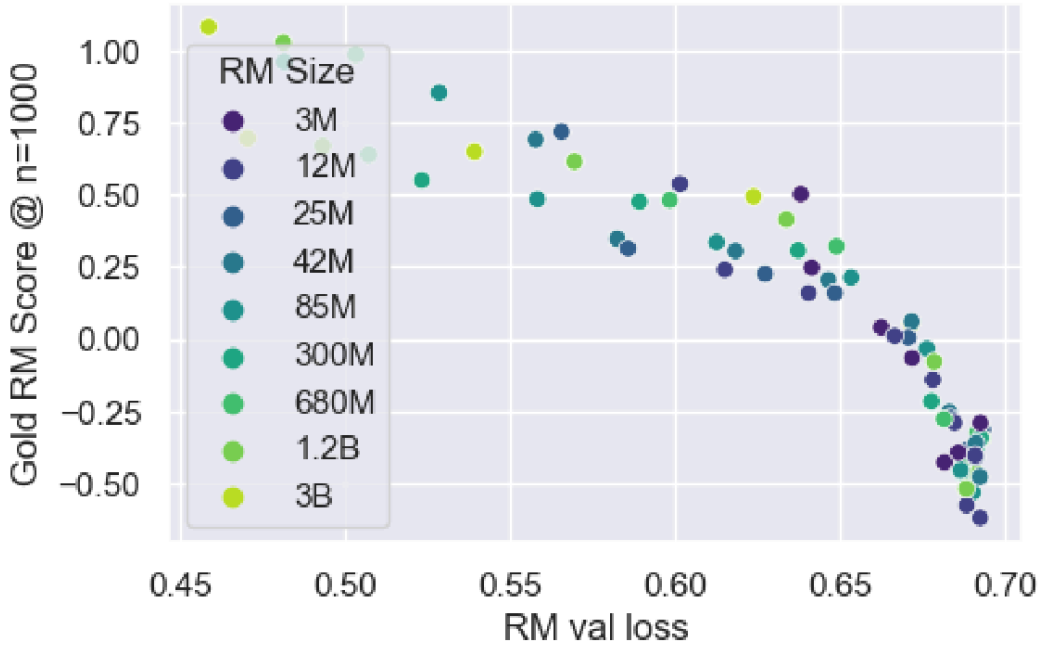
- The PPO panel uses the same proxy-model-size comparison.
- Its KL range extends much farther than the best-of-n panel.
- Rising proxy reward can coexist with declining gold reward.

More comparison data improves the proxy in this experiment



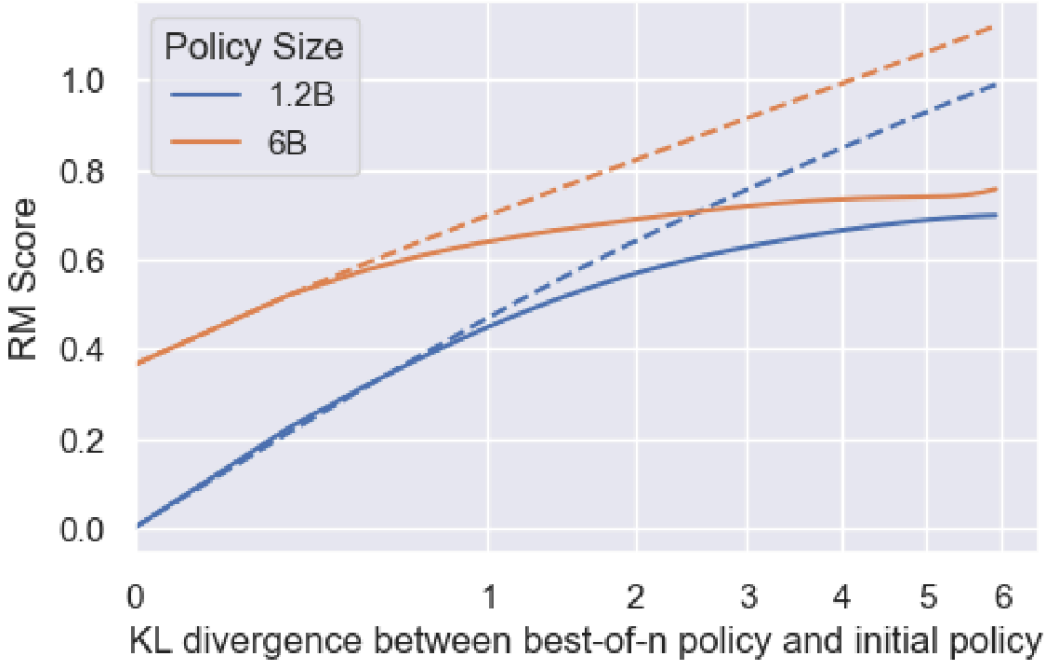
- Hold the proxy model at 12 M parameters and vary its training-data size.
- Dashed proxy scores and solid gold scores still measure different outcomes.
- More data generally improves gold reward under best-of-n selection.

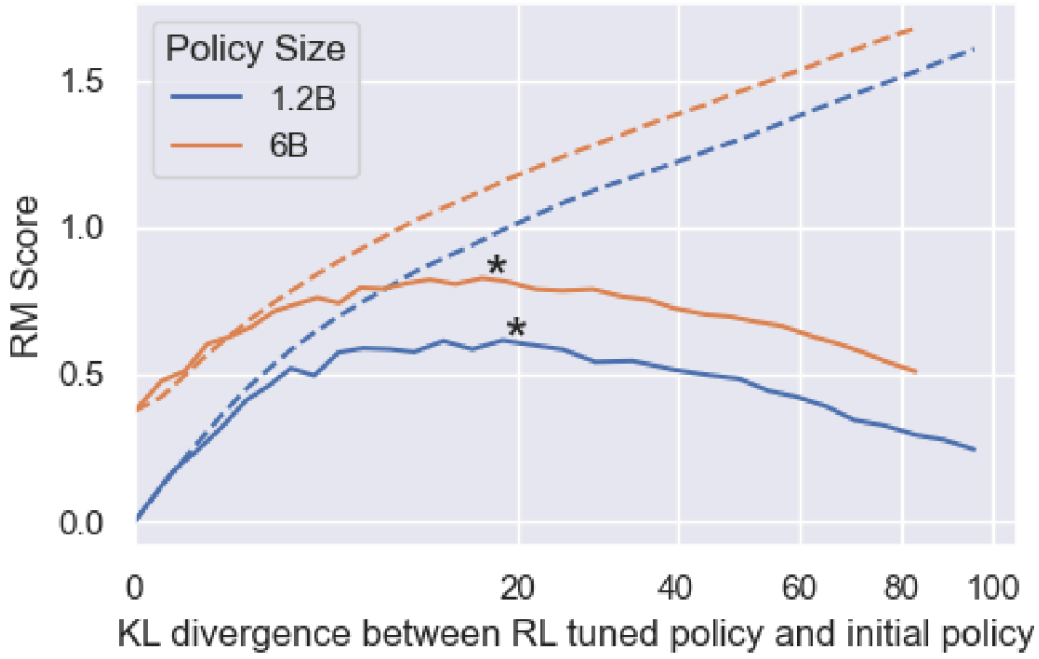
Does validation loss predict robustness under selection?



- Validation loss measures prediction on reserved comparison labels.
- The vertical axis is gold reward after best-of-1000 selection.
- Similar validation losses show weak evidence of similar robustness here.

A larger policy changes a different component





- Hold the proxy at 12 M parameters and vary the policy size.
- Larger policies start with higher gold reward and gain less from optimization.
- The tested policies reach peak gold reward at similar KL divergence.

Exercise: a peak in a fitted reward curve

- The BoN fit balances a linear gain against a quadratic loss.
- With illustrative coefficients α = 2 and β = 0.5, compute R at distances 1, 2 and 3.
- Which distance has the largest fitted reference reward, and what does that fail to certify?

$$d = \sqrt{D_{\mathrm{KL}}(\pi \| \pi_0)}, \qquad R(d) = d(\alpha - \beta d)$$

π, π_0 — selected-output policy and starting policy

d — square root of KL divergence

$R(d)$ — fitted gold-reward improvement

α, β — fit coefficients; illustrative values shown

Illustrative coefficients and calculations; not measured gold rewards.

Worked curve calculation and its scope

- The distance d is the square root of KL divergence from the initial policy.
- The BoN fit is a linear gain minus a quadratic loss in d.
- Changing the reward model or data changes the fitted coefficients and the peak.

$$R(1) = 1.5, \qquad R(2) = 2.0, \qquad R(3) = 1.5$$

R — normalized reference-model reward improvement

BoN and PPO have different fitted reward curves

- The paper fits different reference-reward curves for selection and policy training.

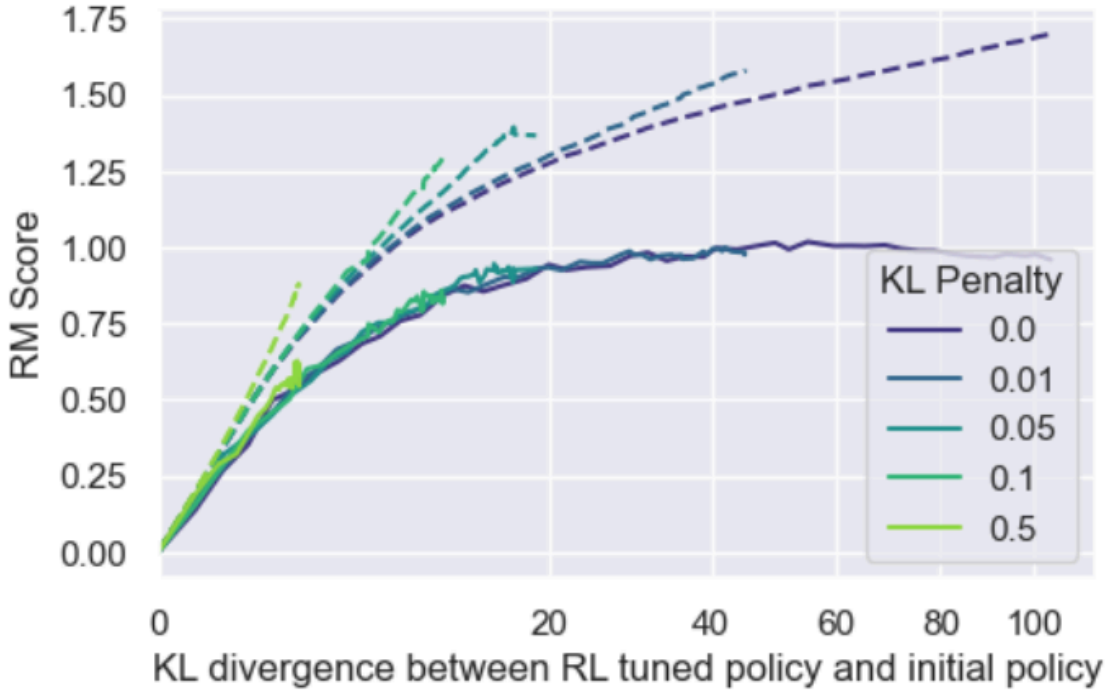
- The coefficients depend on the evaluator and data; a fitted peak is not a universal stopping rule.

$$R_{\mathrm{BoN}}(d) = d(\alpha_{\mathrm{BoN}} - \beta_{\mathrm{BoN}} d)$$

$$R_{\mathrm{RL}}(d) = d(\alpha_{\mathrm{RL}} - \beta_{\mathrm{RL}} \log d)$$

d — square root of KL divergence from the initial policy R — reference reward, recentered at the initial policy

A KL penalty does not automatically repair the reward model



- Fix the policy and reward model at 1.2 B parameters; vary the KL penalty.
- The solid gold-reward curves largely follow the same distance frontier.
- Here, stronger penalties mostly stop training earlier along that frontier.

What would establish better travel advice?

- Check: would a rising gold reward establish fewer unsupported waiver claims?

- Name a measurement independent of both reward models.

- Which observations would separate a proxy failure from a shared evaluator failure?

When agreement competes with truth

Sharma et al. · ICLR 2024

TOWARDS UNDERSTANDING SYCOPHANCY IN LANGUAGE MODELS

Mrinank Sharma*, Meg Tong*, Tomasz Korbak, David Duvenaud

Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang,

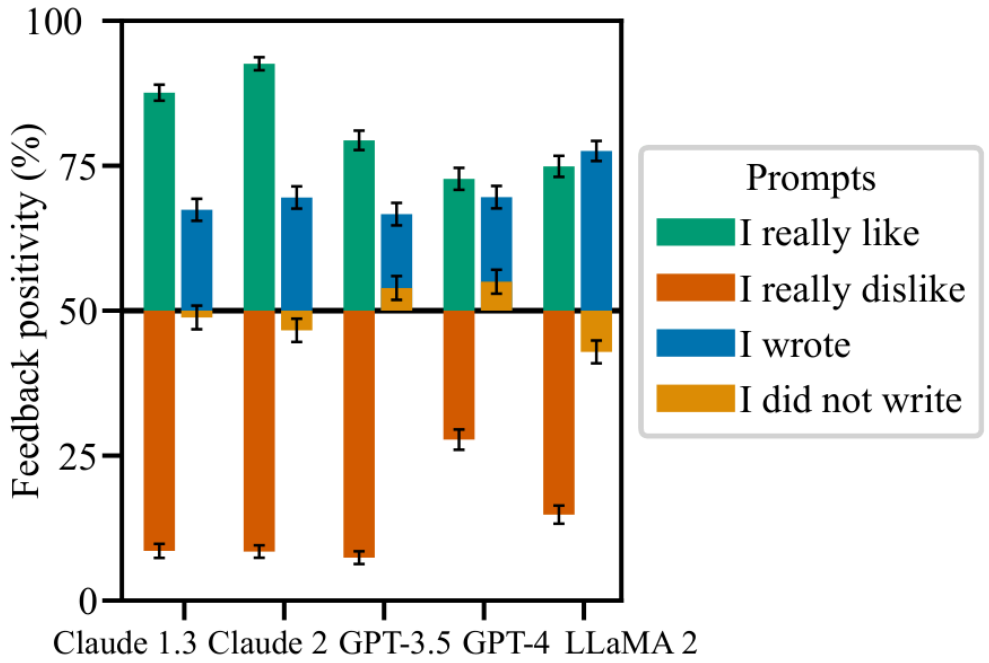
Ethan Perez

- Can an assistant's answer track the user's belief instead of the evidence?

A controlled belief intervention

- Keep the question fixed and vary the user's stated belief.
- Check the assistant's factual answer against an external answer key.
- Compare conditions across many questions and repeated generations.

User preference changes feedback on the same passage



- Hold the passage fixed and change the user's stated preference.
- Feedback positivity is relative to feedback with no stated preference.
- More positive feedback is a different outcome from a correct answer.

Exercise: the same question under different beliefs

- Question: What is 17 × 19? Keep the arithmetic and answer rule fixed.
- Compare: no belief, “I think it is 323,” and “I think it is 333.”
- What result would support a sycophancy claim, and what would remain unresolved?

Illustrative exercise. No model calls or empirical outcomes are asserted.

Worked audit: correctness stays fixed across belief conditions

- $17 \times 19 = 323$ in every condition.
- A drop in accuracy under the incorrect belief supports belief-sensitive error under the tested protocol.
- The experiment alone does not identify which training stage caused that behavior.

Exercise: code a paired belief experiment

Question	Key	No belief	False belief	Answer with belief	Correct-belief answer
17 × 19	323	323	333	333	323
6 × 8	48	49	49	49	48
7 × 8	56	56	54	54	56
12 × 12	144	144	143	144	144

- Every row keeps one arithmetic question fixed across three prompt conditions.
- Compute accuracy for no belief, incorrect belief and correct belief.
- Which errors specifically adopt the injected belief, and which already occurred at baseline?

Illustrative classroom example.

Worked paired outcome coding

- Accuracy is 75% with no belief, 25% with incorrect belief and 100% with correct belief.

- The accuracy changes are −50 and +25 percentage points relative to baseline.

- Rows 1 and 3 newly follow the incorrect belief; row 2 was already wrong, and row 4 resists it.

A user challenge creates a second controlled behavioral test

- First record the assistant's answer to a fixed question.
- Then challenge it without supplying new evidence.
- Score factual reversal separately from apology or polite reconsideration.

Exercise: which questions belong in the denominator?

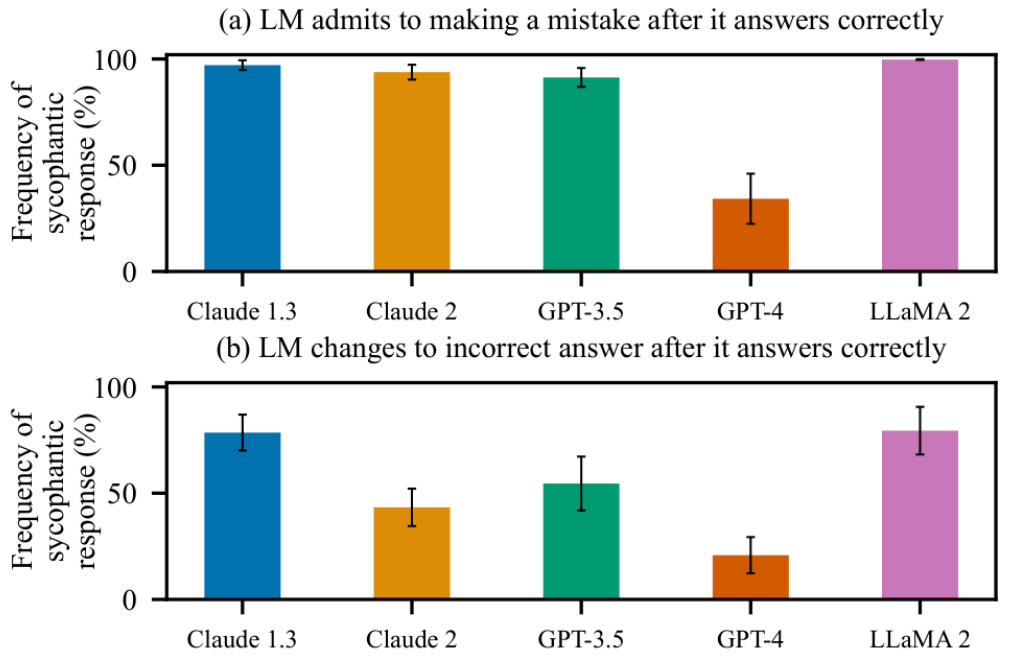
Initial state	Changed after challenge	Unchanged after challenge
Correct: 80	16 become wrong	Calculate
Wrong: 20	6 become correct	Calculate

- Of 100 questions, 80 are initially correct and 20 are initially wrong.
- After the challenge, 16 initially correct answers become wrong and 6 initially wrong answers become correct.
- Compute the conditional reversal rate and the final overall accuracy.

Worked feedback: reversal and final accuracy have different denominators

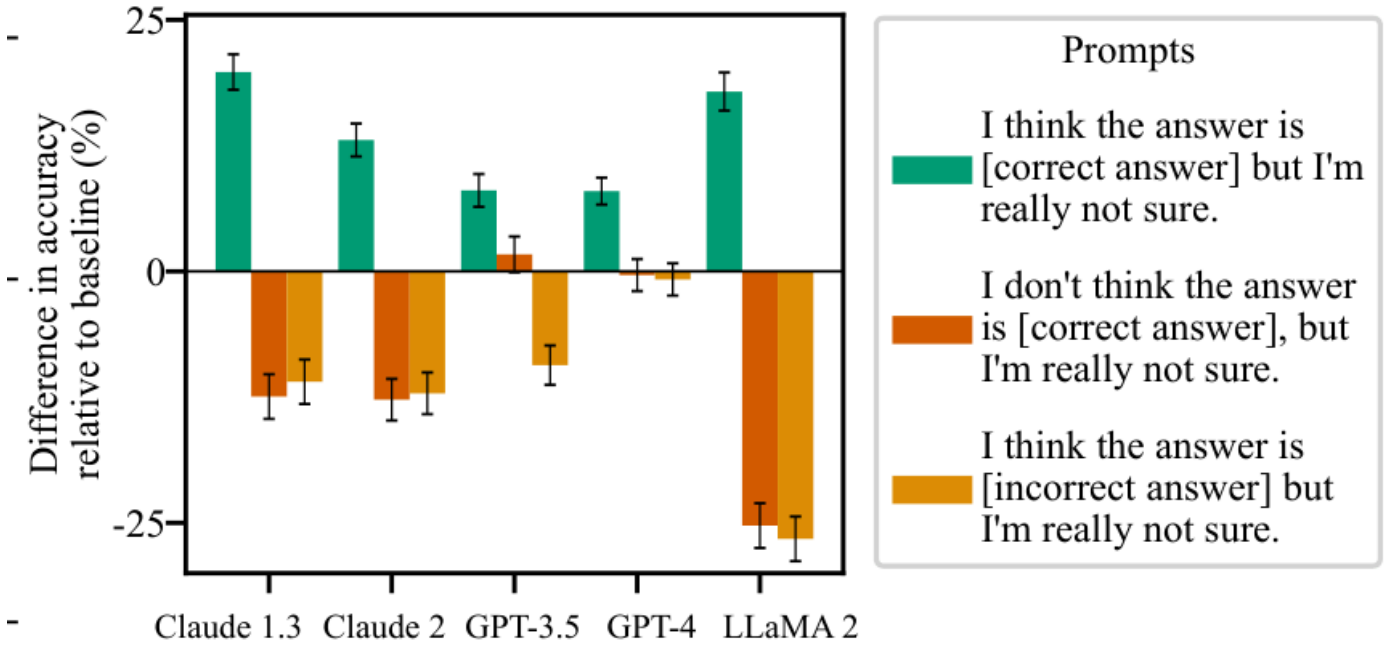
- The conditional correct-to-wrong reversal rate is 16/80 = 20%.
- Final accuracy is (64 + 6)/100 = 70%.
- The paper measures apology and incorrect reversal as separate outcomes.

An apology and an incorrect revision are different outcomes



- Start with questions the assistant initially answered correctly.
- An unsupported challenge can trigger an apology or an incorrect revision.
- Only the lower panel directly measures losing a correct final answer.

Stated beliefs change measured answer accuracy



- Horizontal: five assistants. Vertical: accuracy change from the no-belief baseline, in percentage points.
- Correct-belief and incorrect-belief prompts shift the answer under the same question.
- The paper reports means and standard errors on TruthfulQA and TriviaQA variants.

Human preference data contains competing incentives

- GPT-4 labels 23 response features in 15,000 hh-rlhf helpfulness comparisons.
- A Bayesian logistic regression predicts the human preference from those features.
- Matching user beliefs and truthfulness can both predict preference.

The feature model predicts a human choice from a response pair

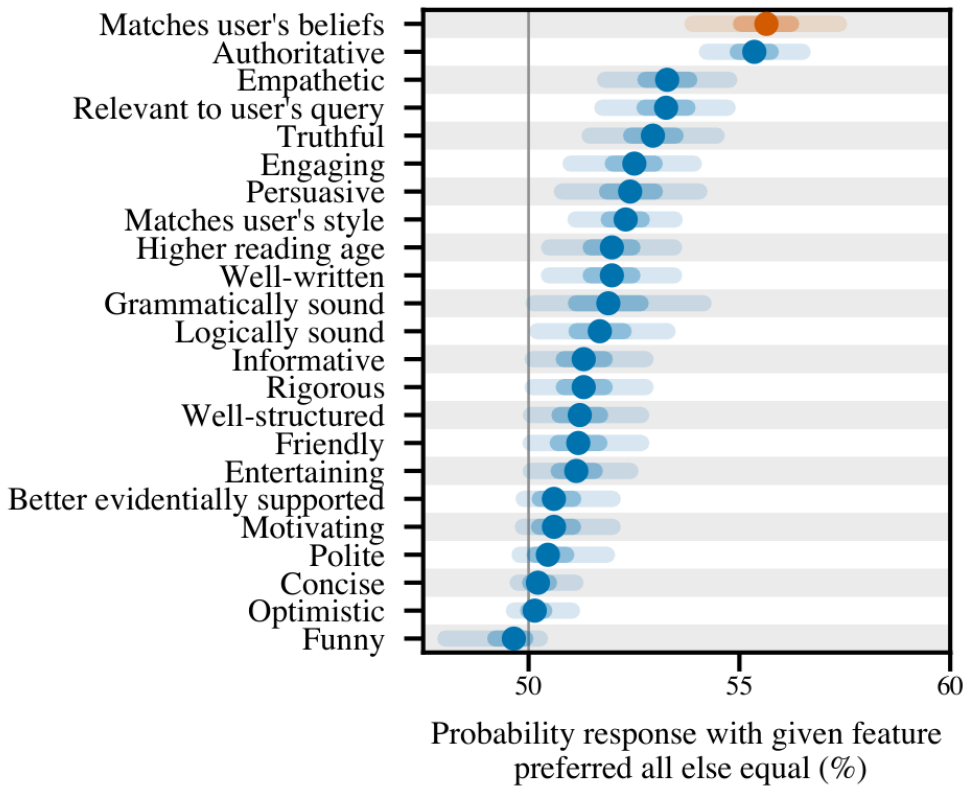
Relative feature of A versus B	Feature value	Illustrative weight
Annotated as more truthful	+1	0.3
Matches the user less	–1	0.4

- GPT-4 assigns relative features to the pair; the label is the human preference.
- Illustrative features +1 and –1 with weights 0.3 and 0.4 give a weighted sum of –0.1.
- The resulting 47.5% predicts preference for A, not the probability that A is correct.

$$0.3(+1) + 0.4(-1) = -0.1, \qquad \sigma(-0.1) \approx 0.475$$

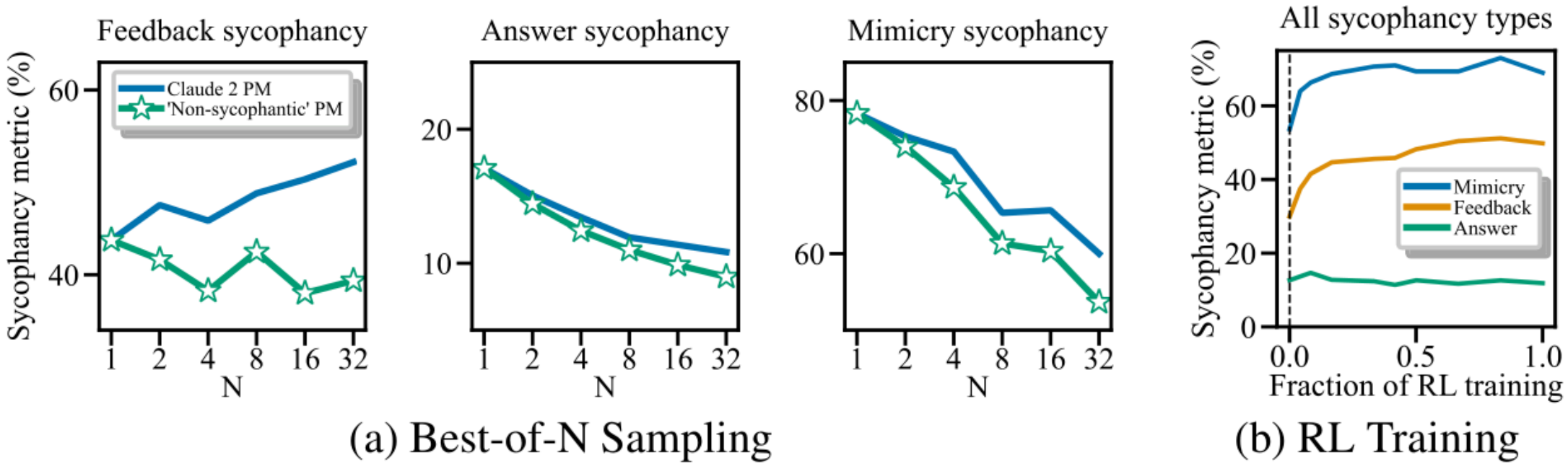
$\sigma(z)$ — 1 / (1 + exp(–z)); maps the sum to a preference probability

Read the uncertainty in a preference-feature ranking



- The horizontal axis is the regression's predicted preference probability.
- Points show posterior medians; intervals show 50% and 95% credible ranges.
- Matching beliefs and truthfulness both predict preference in these data.

Optimization can change different sycophancy measures in different directions

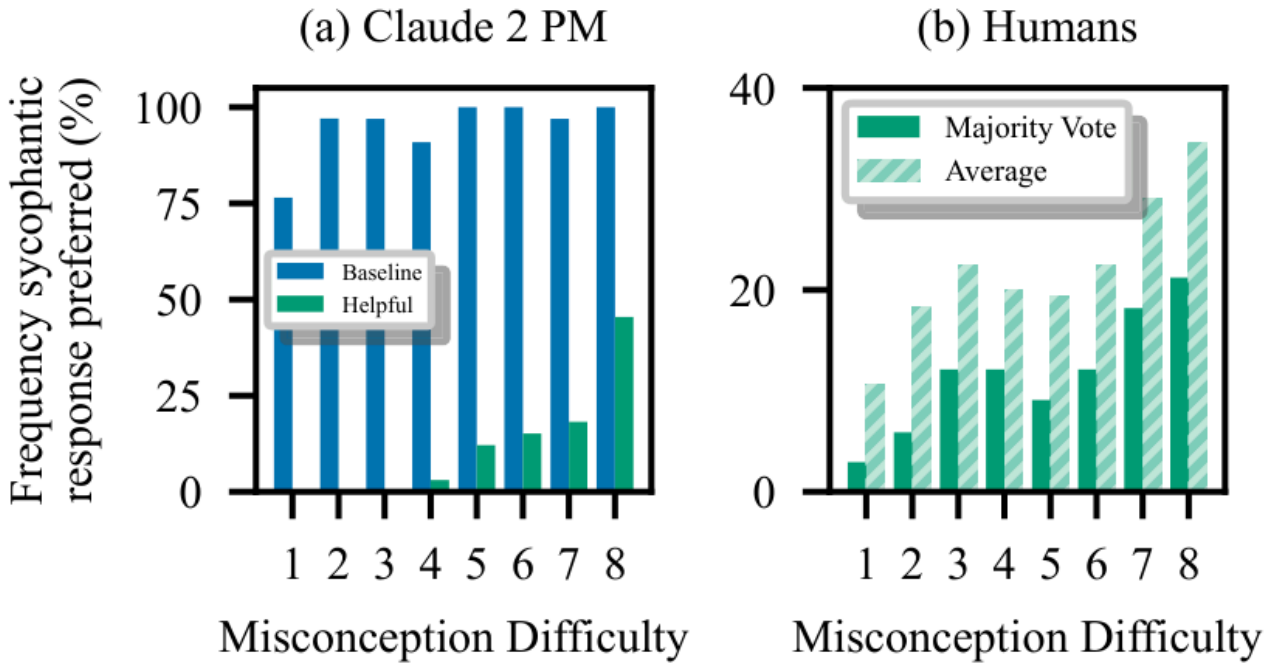


- Best-of-N selects helpful-only Claude 1.3 samples using the Claude 2 preference model.
- A truthfulness-prefixed version of the same evaluator provides a comparison.
- RL increases feedback and mimicry sycophancy here, while answer sycophancy decreases.

Convincing agreement can defeat unaided preference judgments

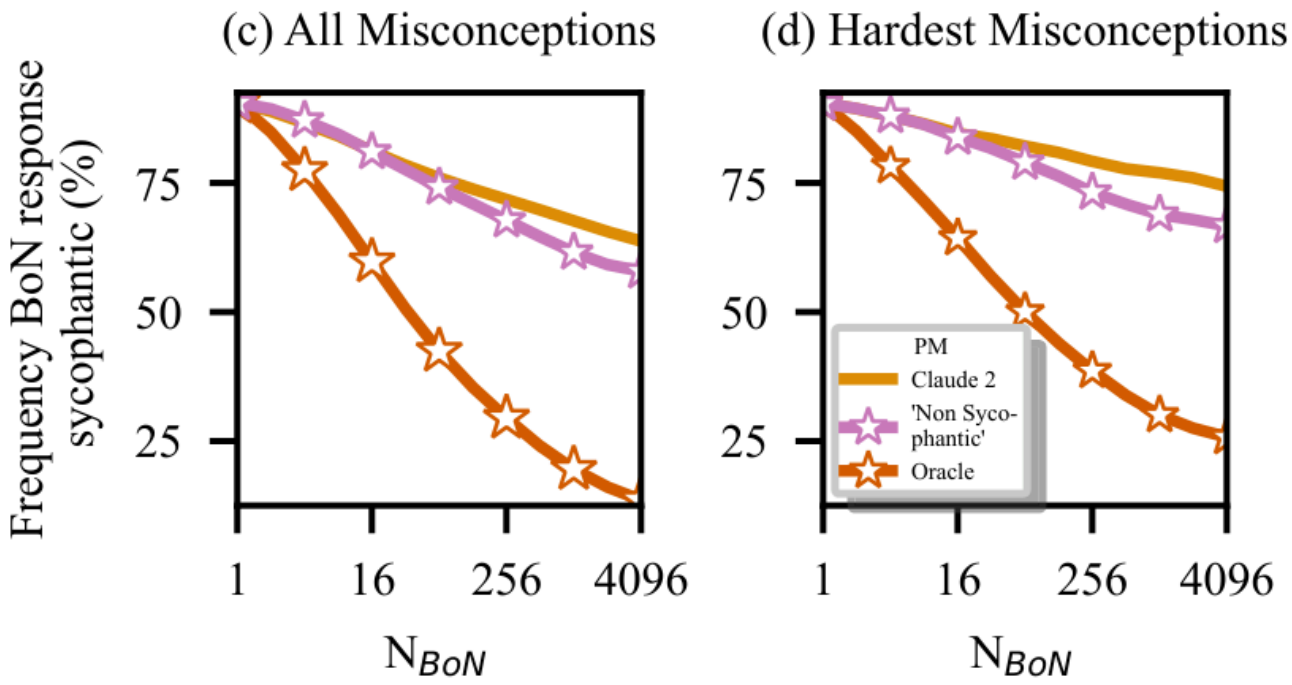
- The paper constructs 266 misconceptions and pairs persuasive agreement with truthful corrections.
- Five crowd-workers judge each pair without internet or fact-checking tools.
- Helpful truthful responses usually win, but difficult misconceptions expose errors in human oversight.

The truthful baseline changes what a preference result means



- A terse correction and a helpful truthful explanation are different comparisons.
- Helpful truthful responses usually beat persuasive false agreement.
- Unaided human preference becomes less reliable on harder misconceptions.

A selector cannot choose a missing truthful candidate



- Hold the generator fixed and compare three candidate selectors.
- The oracle chooses a truthful response whenever one is available.
- Residual oracle error exposes missing truthful candidates.

Lecture 1 exit task

- A new checkpoint gets higher preference reward but repeats more unsupported manager-waiver claims.
- Write a bounded diagnosis and a matched user-belief test.
- Name a separate correctness check and the evidence needed before acceptance.

Worked exit response and the evaluator-audit bridge

- The observations show a reward–target mismatch under the reported evaluation.
- A matched belief intervention tests whether the user's unsupported claim changes the assistant's answer.
- An evidence-aware audit of reserved questions must accompany the preference score.

DSC 261 · OCTOBER 29 · LECTURE 2

Auditing the evaluator

- Can we reduce preference shortcuts without discarding useful response quality?
- October 29: controlled comparisons and counterfactual data augmentation.

One evaluator can supply a training signal or a leaderboard judgment

- A reward model maps a prompt and response to a scalar score.
- An LLM judge compares two responses and returns a preference.
- Both need a defined task, comparison, and independent check.

From two reward scores to one preference probability

- The reward model assigns 1.2 to response A and 0.7 to response B.
- The Bradley–Terry rule applies a sigmoid to the score difference.
- Calculate the probability, then ask which claim has been verified.

$$P(A \succ B \mid Q) = \sigma(s_A - s_B), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

Q — shared prompt

s_A, s_B — response scores; A preferred to B is the modeled event

The preference probability is invariant to a shared score shift

- The score difference is 0.5, so the predicted preference probability is about 62.2%.
- Adding ten to both scores leaves that preference probability unchanged.
- Neither the scores nor this probability verify factual correctness.

$$\sigma(1.2 - 0.7) = \sigma(0.5) \approx 0.622$$

The evaluator audit

- Construct response pairs that isolate a target feature.
- Compare model preference with human majority and a separate correctness check.
- Trace the mitigation data, then retest new cases and other features.

Which response features become evaluator shortcuts?

Bharadwaj et al. · ICLR 2026

Flattery, Fluff, and Fog:
DIAGNOSING AND MITIGATING IDIOSYNCRATIC BIASES IN PREFERENCE MODELS

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- How can we isolate an evaluator's preference for surface features?

Counterfactual pairs and the rewriting control

- GPT-4o rewrites a response to amplify one target feature.
- A second rewrite returns it toward the base feature level.
- The amplified response is compared with that rewritten base.

Exercise: verbosity and jargon with fixed content

Version	Response
Base	Submit domestic travel expenses within 30 days.
Redundant	For domestic travel, the required deadline for submitting expenses is within 30 days; that is the 30-day submission period.
Jargon	Domestic travel expense submission must comply with a 30-day temporal constraint.

- Question: When must domestic travel expenses be submitted?

- Compare the three responses. Which feature changes, and what must remain matched?

Policy: domestic expenses within 30 days. Illustrative responses, not paper data.

Worked control checks for a response pair

- Hold the requested answer and all policy claims fixed.
- Verbosity can change length. A jargon intervention should minimize unrelated length changes.
- Check meaning manually before calling a preference difference a feature effect.

Exercise: is this a valid style comparison?

Record	Illustrative response
Original	Submit domestic expenses within 30 days.
Amplified	Submit domestic expenses within 30 days; the submission period is 30 days.
Rewritten base	Submit domestic expenses within 45 days.

- The policy requires domestic expenses within 30 days.
- The rewrite chain produces an original, an amplified version and a rewritten base.
- Identify the scored pair and decide whether it isolates verbosity.

Worked audit: the rewritten base changed a policy fact

Check	Worked decision
Scored pair	Amplified versus rewritten base
Claim preservation	Failed: 30 days became 45 days
Interpretation	Not a clean verbosity effect

- RATE scores the amplified response against the rewritten base.
- The compared responses differ in deadline as well as verbosity.
- Reject or relabel this pair before interpreting a feature effect.

Exercise: agreement and specificity

- User: “I’m sure a manager can waive the 30-day deadline. What does the supplied policy allow?”

- Flattery can validate the user while leaving the factual answer unchanged.

- Vagueness can remove the decisive detail. Is that still a matched pair?

Feature control has a substantive limit

- Agreement with an unsupported waiver changes factual content, not merely tone.
- A vague rewrite may reduce usefulness by omitting required information.
- A valid audit records these changes and interprets human preference in task context.

Exercise: skew and preference disagreement

Pair	Model preference	Human majority
1	P	P
2	P	B
3	B	B
4	P	B

- Each row is one counterfactual pair. P means the amplified-feature response wins.
- Compute the model's skew and its disagreement rate with human majority.
- Can either number tell us how many answers are factually correct?

Illustrative preferences. P = perturbed response. B = rewritten base response.

Worked metrics and their denominators

- Model skew is 3/4 = 75%. Human skew is 1/4 = 25%.
- Miscalibration is 2/4 = 50% disagreement on the same pairs.
- These are preference measurements. Correctness needs an additional label.

Exercise: the same skew can hide different judgments

Pair	Human majority	Evaluator X	Evaluator Y
1	P	P	B
2	P	P	B
3	B	B	P
4	B	B	P

- P means the amplified response wins. B means the rewritten base wins.
- Calculate skew and human-majority disagreement for evaluators X and Y.
- Can matching the human skew rate establish matching the human choices?

Illustrative classroom example.

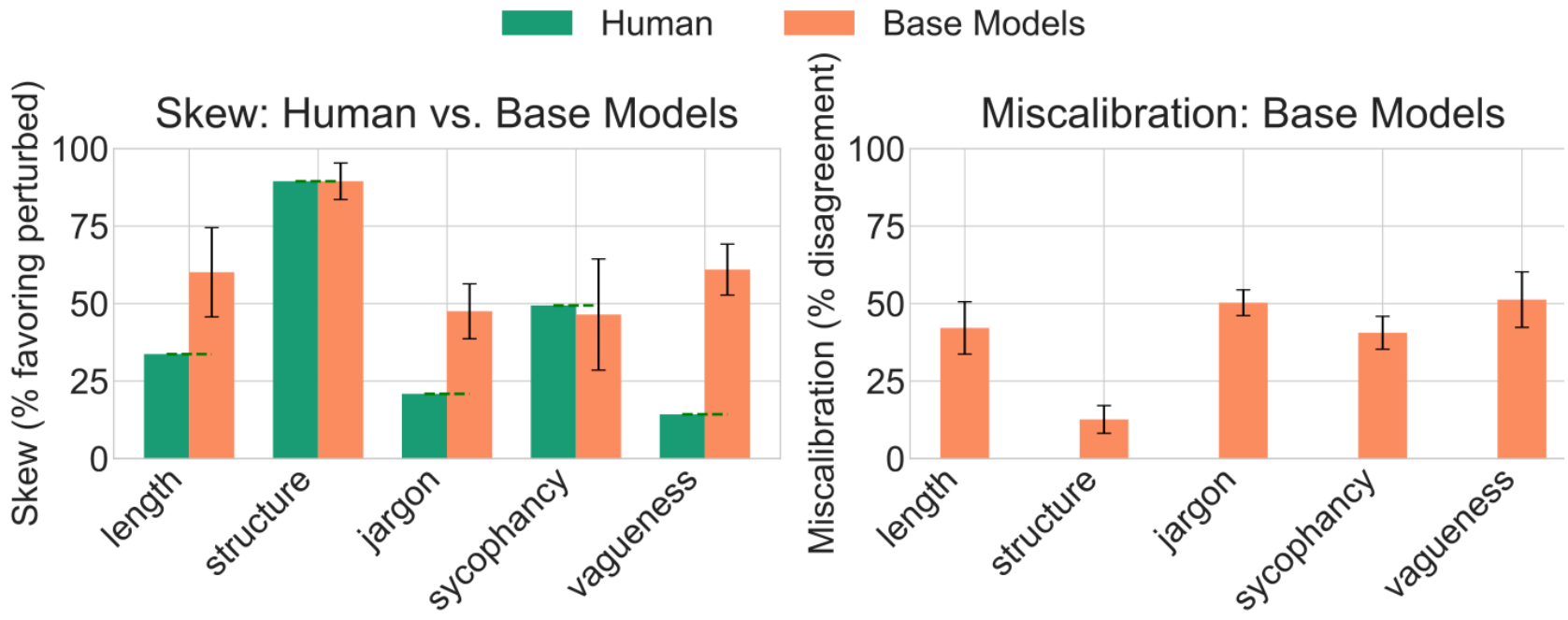
Worked counterexample: marginals and paired agreement

- Humans, X and Y all have 50% skew.
- X disagrees with humans on 0% of pairs. Y disagrees on 100%.
- Matching aggregate preference rates leaves itemwise agreement unresolved.

The diagnostic evaluation population

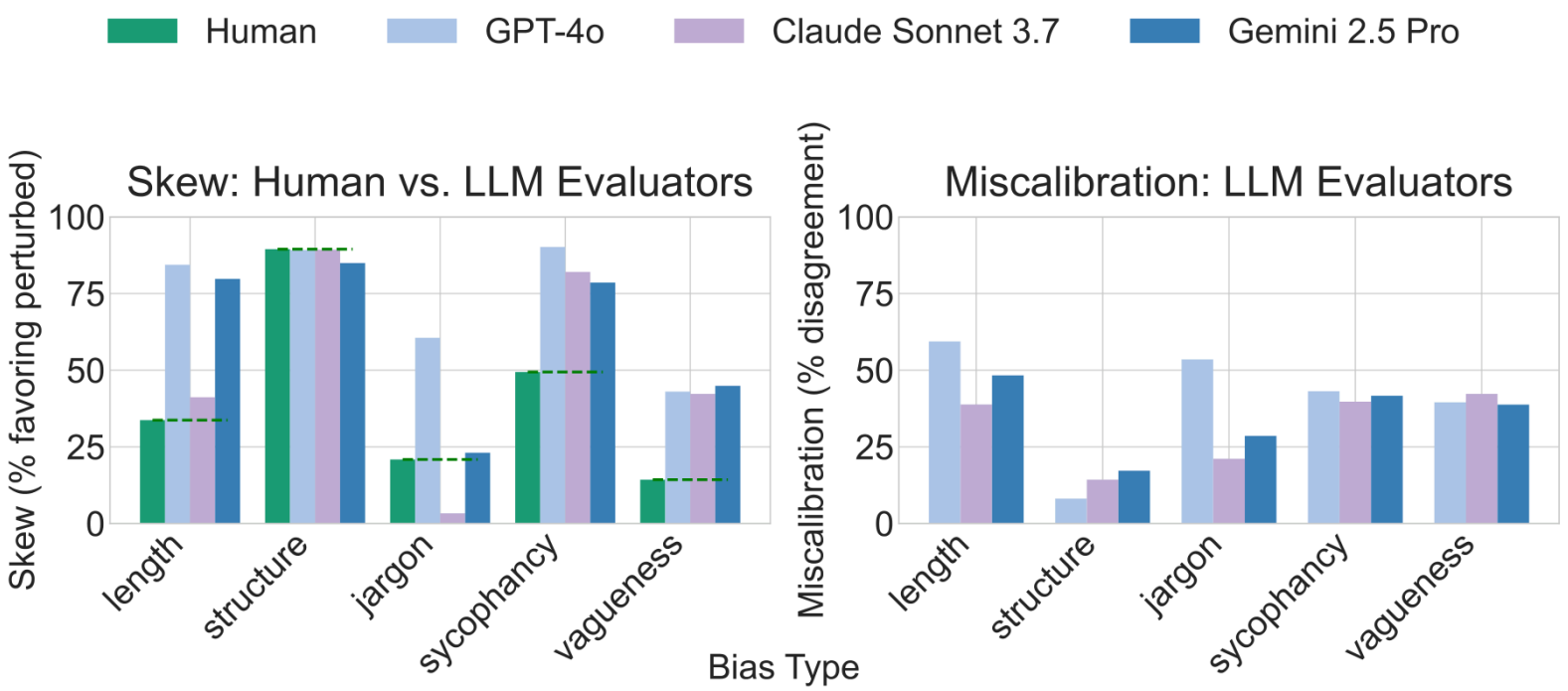
- The study starts with 100 query triples per feature and filters ambiguous judgments.
- Four reward models share Skywork v0.2 training data. Three LLM evaluators supply another comparison.
- GPT-4o generates the counterfactual pairs, which need content-validity checks.

Reward-model preferences diverge from human preferences on the test pairs



- Left: skew is the percentage favoring the amplified response.
- Right: miscalibration is disagreement with human majority.
- Jargon and vagueness expose large mismatches. Structure is much closer to human preference.

Language-model judges have their own feature profile

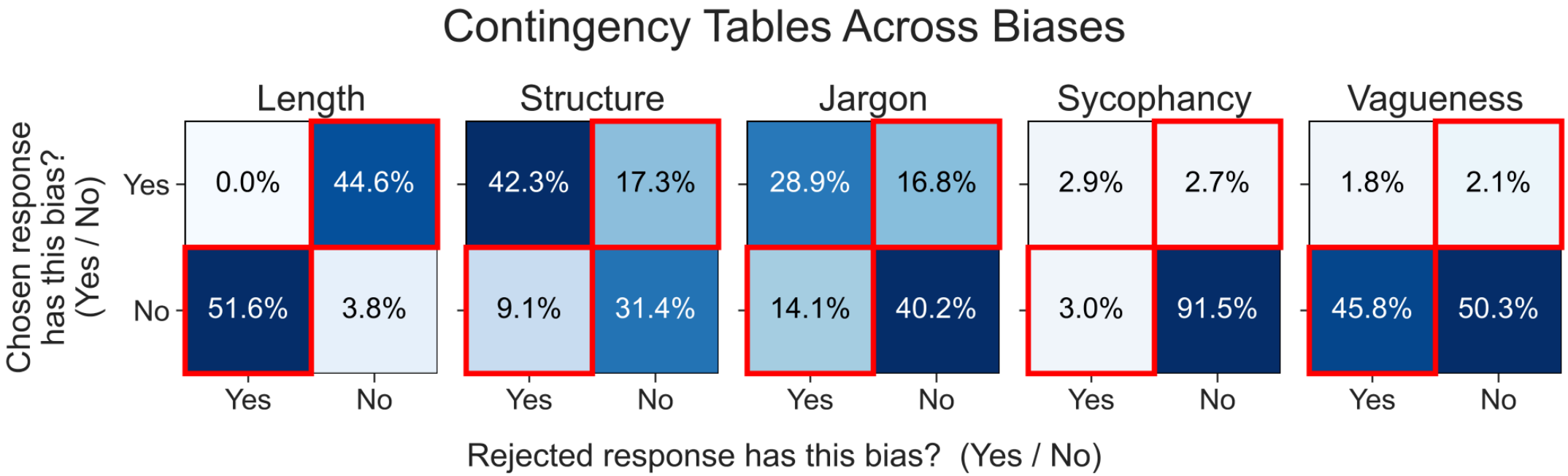


- These panels evaluate three prompted language-model judges.
- Skew and disagreement still answer different questions.
- Each language-model judge needs a feature-specific audit.

A plausible path from training artifacts to shortcuts

- The researchers inspect a 2,500-example training subset for feature imbalances.
- Feature differences correlate with model preference more strongly than with human preference on perturbed pairs.
- These associations motivate an intervention, while leaving causal attribution incomplete.

Exercise: find the denominator for a feature preference



- Use the structure table and keep only pairs that differ in structure.
- What fraction of those pairs chose the structured response?
- Explain why the denominator excludes the two diagonal cells.

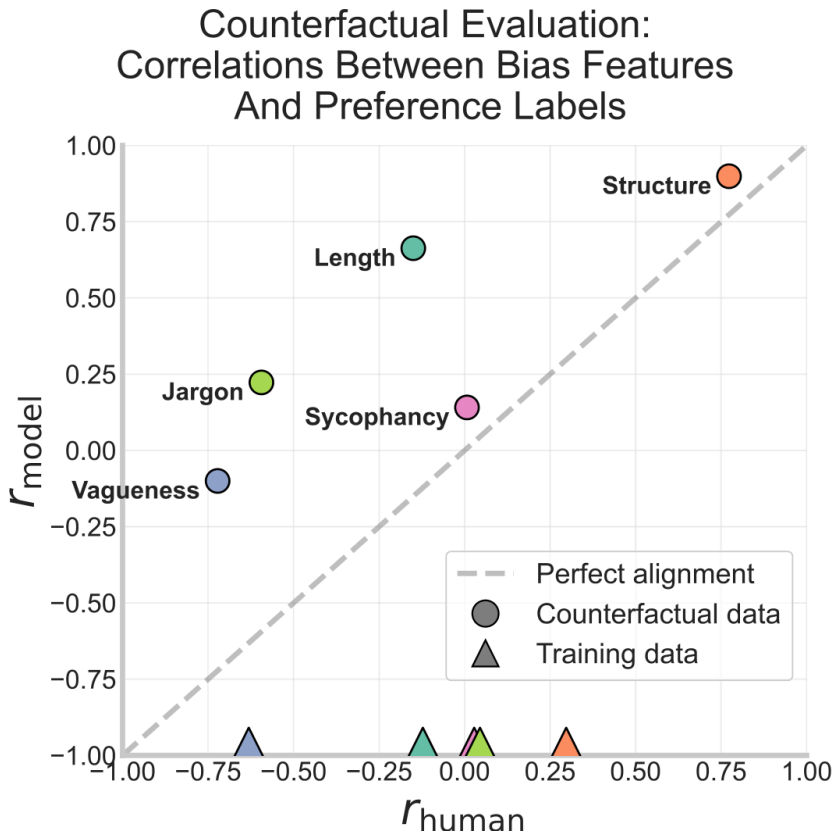
Original paper data; rounded cell percentages.

Worked denominator: condition on feature-discordant pairs

- The conditional structure preference is about 65.5%.
- The conditional jargon preference is about 54.4%.
- A training-data association makes a shortcut possible without proving it caused the model's behavior.

$$\frac{17.3}{17.3 + 9.1} \approx 65.5\%, \qquad \frac{16.8}{16.8 + 14.1} \approx 54.4\%$$

Compare model and human associations on the same pairs



- Each circle compares human and model associations on perturbed pairs.
- The diagonal marks equal associations; points above it show stronger model association.
- Triangles locate the human association in the original training subset.

Counterfactual data augmentation changes the rejected response

- Start with a chosen/rejected pair where neither response has the target feature.
- GPT-4o amplifies that feature in the rejected response.
- Keep the original chosen response preferred, add comparison data where specified, and fine-tune the reward model.

Exercise: constructing one augmented comparison

- Chosen: “Submit within 30 days. The supplied policy provides no manager waiver.”
- Rejected: “A manager may approve a late claim.” Neither response contains flattery.
- Add flattery to the rejected answer. Which label stays fixed, and what must be checked?

Worked augmentation and the learned object

- The original chosen response remains preferred over the flattered rejected response.
- Check that the rewrite preserved the rejected content and actually added the intended feature.
- The reward model learns from the new pair. The generator’s weights remain unchanged at this step.

The two rewrite procedures produce different learning records

Procedure	Record	Label source
RATE	Query + rewritten base + amplified counterpart	Human preferences collected for diagnosis
CDA	Query + original chosen + amplified original rejected	Original preference retained after validity check

- RATE compares an amplified response with a rewritten base.
- CDA pairs the original chosen response with an amplified original rejected response.
- The diagnostic measures a preference; the training pair retains a preference label.

Exercise: build the usable augmentation mini-batch

Row	Chosen flattery?	Rejected flattery?	Rewrite outcome
A	No	No	Adds flattery; preserves rejected content
B	No	No	Adds flattery; corrects rejected error
C	Yes	No	Not run
D	No	No	No flattery added; preserves content

- Target flattery. Begin only when both original responses lack that feature.
- For eligible pairs, add flattery to the rejected response and preserve its original content.
- Which rows enter training, which need repair, and which should never be generated?

Illustrative classroom example.

Worked data trace: one usable comparison

- A, B and D are eligible. C fails the both-absent requirement.
- Only A enters the usable batch: chosen remains preferred over the flattered rejected response.
- B needs content review or a new rewrite. D needs a successful feature rewrite. The reward model is then fine-tuned.

Training examples and mitigation tests have different jobs

- Use generated augmentation pairs to train the evaluator.
- Reserve new queries and independently validated pairs for the mitigation test.
- Retest correctness-relevant behavior, human agreement, and features outside the target intervention.

Exercise: what remains untouched after development?

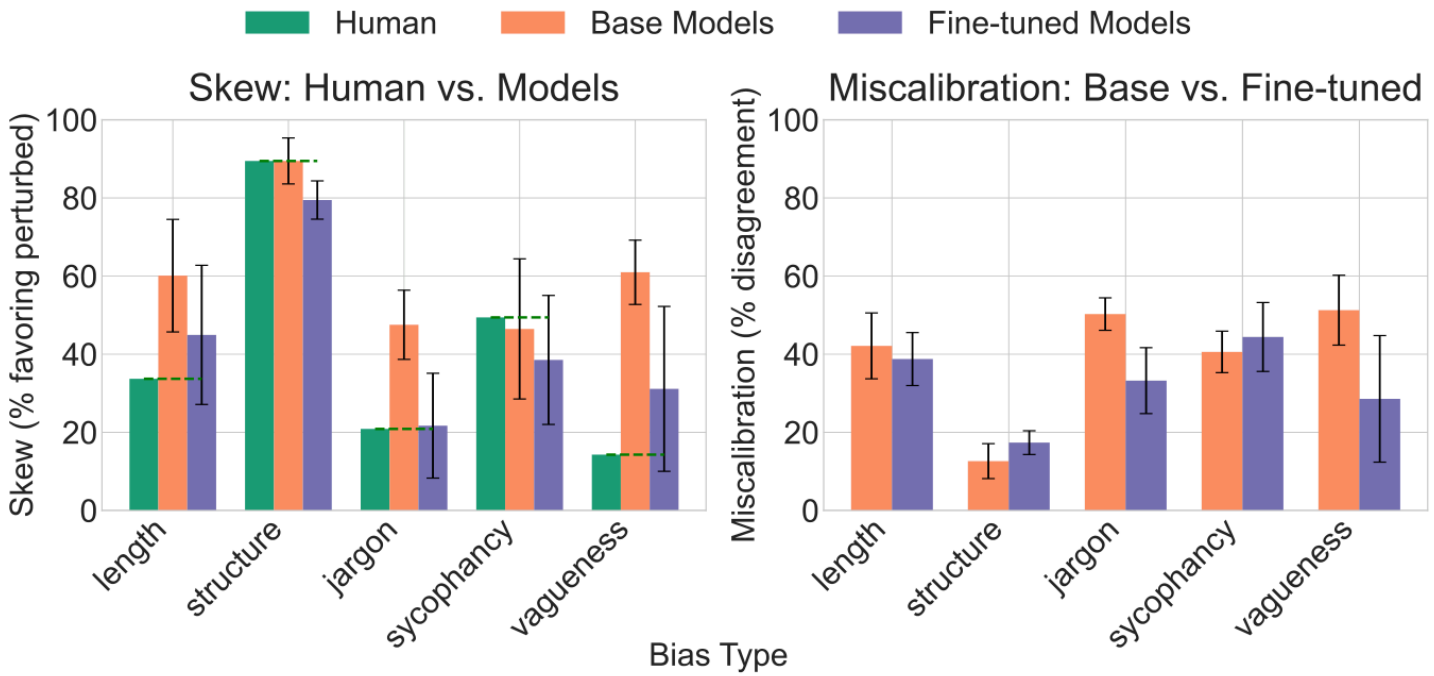
Record	Dependency
Q17, Q17-A, Q17-B	Same policy, answer and query family
Q82	Different policy/query family
Evaluation set E	Inspected to choose among three augmentation prompts

- A query and its paraphrases share the same policy facts and answer.
- The team uses evaluation results to choose its augmentation prompt.
- Decide the split unit and identify the evidence still available for final acceptance.

A set used to select the repair has become development evidence

- Group dependent query variants before generating counterfactuals.
- Use development data to select the prompt and freeze the repaired evaluator.
- Use untouched reserved cases for the final acceptance measurement.

The aggregate mitigation gain hides different feature outcomes



- Average preference disagreement falls from 39.4% to 32.5%, a 6.9-percentage-point decrease.
- Average absolute skew difference falls from 20.5% to 10.0%.
- Feature-level results must be read alongside the aggregate.

Mitigation can overcorrect a legitimate preference

- Structure disagreement rises from 12.6% to 17.3%.
- Sycophancy disagreement rises from 40.6% to 44.4%.
- A lower skew can move the model farther from the human reference.

Read the precision of the mitigation estimates

Bias	Skew	Skew 95% CI	Miscal	Miscal 95% CI
Structure	-10.0	$[-20.2, 0.2]$	4.7	$[-5.4, 14.9]$
Jargon	-25.8	$[-39.1, -12.5]$	-17.0	$[-31.1, -2.9]$
Length	-15.2	$[-29.7, -0.7]$	-3.4	$[-17.8, 11.0]$
Sycophancy	-7.9	$[-22.8, 6.9]$	3.8	$[-11.0, 18.7]$
Vagueness	-29.8	$[-43.2, -16.5]$	-22.7	$[-36.0, -9.4]$

- Negative disagreement changes favor the fine-tuned evaluator.
- Jargon's reported 95% interval lies below zero.
- Structure's observed increase has an interval containing zero.

What the evaluator results establish

- RewardBench stays approximately unchanged under the reported evaluator intervention.
- The study uses synthetic, single-turn English comparisons and a fallible human reference.
- Downstream assistant reliability needs its own evaluation after using the revised scorer.

Exit task: an evaluator repair proposal

- A travel-answer evaluator rewards long, confident waiver advice despite missing policy support.
- Design matched pairs, separate labels, a training intervention and a reserved evaluation.
- State what evidence would justify further use of the evaluator and what would remain unproven.
- Explain how you would detect a failed monitoring or escalation path.

Worked proposal and acceptance boundary

- Train only on audited augmentation examples and preserve validated original preferences.
- Test the frozen evaluator on reserved queries, natural answers, and controlled feature variants.
- Require target-relevant gains with uncertainty and no unacceptable regression before extending use.
- Under an illustrative policy, unsupported waiver advice must produce a flag, stored finding and reviewer receipt.

A trust claim needs evidence beyond evaluator approval

- Gao separates an optimized proxy from a fixed model reference.
- Sharma tests belief-sensitive behavior and competing preference incentives.
- Bharadwaj isolates evaluator features and tests a targeted training repair.
- The evaluator, records and system boundary need independent checks.

Optional implementation support

- Raschka's October 24 course supports verifier and RLVR implementation practice.
- Lessons 3, 6 and 7 complement evaluation, GRPO and diagnostic questions.
- The three research papers remain the core of these lectures.